

Class Test

Physics-I (EEE & CSE)

Date: 10.01.2021

Time: 40 minutes

Total Marks: 20

[Answer any 4 (four) questions out of 6 Six]

1. Define: Vibration & Force Constant. Write down the properties of SHO. 5
2. Construct the formula for displacement, velocity and acceleration of an object in SHO. 5
3. Investigate the simple harmonic oscillator and indicate the properties of it at different position. 5
4. If a particle of mass 0.05kg executing simple harmonic motion with frequency 50Hz and if its amplitude and maximum velocity are 0.004m and 0.2 ms^{-1} respectively. Calculate its average kinetic energy and the time period of motion. 5
5. Define: Reduced Mass and Write a short note on “Lissajous Figures”. 5
6. Two bodies are connected by a massless spring of force constant 1.5dynes/cm. If the reduced mass of the system be $2g$ and one of the bodies having mass of $10g$, then calculate the time period of the oscillation and the mass of other body connected to the system 5